

[Phosphorus: Fact Sheet for Consumers \(original version\)](#)

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What is phosphorus and what does it do?

Phosphorus is a mineral contained in each cell in our body. Most phosphorus is in the bones and teeth, and some is in your genes. Your body needs phosphorus to make energy and to carry out many important chemical processes.

How much phosphorus do I need?

How much phosphorus you need depends on your age. Average daily recommended amounts are listed below in milligrams (mg).

Life Stage	Recommended Amount
Birth to 6 months	100 mg
Infants 7–12 months	275 mg
Children 1–3 years	460 mg
Children 4–8 years	500 mg
Children 9–13 years	1,250 mg
Teens 14–18 years	1,250 mg
Pregnant and breastfeeding teens	1,250 mg
Adults 19 years and older	700 mg
Pregnant and breastfeeding women	700 mg

What foods provide phosphorus?

Phosphorus is naturally present in many foods. You can get recommended amounts of phosphorus by eating a variety of foods, including the following:

- Dairy products, such as yogurt, milk, and cheese
- Grain products, such as bread, tortillas, brown rice, and oatmeal
- Meats, poultry, fish, and eggs
- Nuts and seeds, such as cashews and sesame seeds

- Legumes, such as lentils, kidney beans, and peas
- Vegetables, such as potatoes and asparagus

Also, many processed foods have additives that contain phosphorus. These additives include phosphoric acid, sodium phosphate, and sodium polyphosphate.

What kinds of phosphorus dietary supplements are available?

Phosphorus is available in a few multivitamin/mineral supplements and some other dietary supplements.

Phosphorus in dietary supplements is often in the form of dipotassium phosphate, disodium phosphate, phosphatidylcholine, or phosphatidylserine. Research hasn't shown that any form of supplemental phosphorus is better than the others.

Am I getting enough phosphorus?

Most people in the United States get more phosphorus than they need from the foods they eat. However, some people are more likely than others to have trouble getting enough phosphorus:

- Premature babies
- People with certain rare genetic disorders
- People with severe malnutrition

What happens if I don't get enough phosphorus?

Phosphorus deficiency is rare in the United States. A phosphorus deficiency can cause loss of appetite, anemia (low red blood cell counts), muscle weakness, coordination problems, bone pain, soft and deformed bones, a higher risk of infection, a feeling of burning or prickling in the skin, and confusion.

What are some effects of phosphorus on health?

Scientists are studying phosphorus to better understand how it affects health. Here are two examples of what this research has shown.

Chronic kidney disease

In severe chronic kidney disease, the kidneys stop working properly and can't get rid of excess phosphorus. Phosphorus then builds up in the blood and can affect bone health and worsen kidney disease, and it might increase the risk of death. Consuming less phosphorus and eating more foods containing calcium might help prevent the side effects of high phosphorus levels in people with severe chronic kidney disease.

Cardiovascular disease

Some studies show that high blood levels of phosphorus might increase the risk of an irregular heartbeat and of death from heart disease. However, other studies haven't found a link between phosphorus levels and heart disease risk. More research is needed to understand whether limiting the amount of phosphorus in a person's diet has any effect on their risk of heart and blood vessel disease.

Can phosphorus be harmful?

High phosphorus intakes seldom cause problems in healthy people. However, you shouldn't get more phosphorus than the upper limits from food, beverages, and dietary supplements unless your health care provider recommends doing this.

The daily upper limits for phosphorus include intakes from all sources—food, beverages, and supplements—and are listed below in milligrams (mg).

Ages	Upper Limit
Birth to 6 months	Not established
Infants 7–12 months	Not established
Children 1–3 years	3,000 mg
Children 4–8 years	3,000 mg
Children 9–13 years	4,000 mg
Teens 14–18 years	4,000 mg
Adults 19–70 years	4,000 mg
Adults 71+ years	3,000 mg
Pregnant teens and women	3,500 mg
Breastfeeding teens and women	4,000 mg

Does phosphorus interact with medications or other dietary supplements?

Yes, phosphorus can interact with medicines that you take, and some medicines can affect phosphorus levels in your body. Here are two examples:

- Antacids containing aluminum hydroxide or calcium carbonate can reduce the amount of phosphorus your body absorbs. Examples of these antacids are Maalox, Rulox, Roloids, and Tums. Using these antacids for 3 months or longer can lead to low levels of phosphorus.
- Some laxatives, such as Fleet Prep Kit #1, contain sodium phosphate and can increase phosphorus levels. Taking more than the recommended dose of these laxatives can be dangerous, especially if you are dehydrated or have kidney disease or heart disease.

Phosphorus and healthful eating

People should get most of their nutrients from food and beverages, according to the federal government's *Dietary Guidelines for Americans*. Foods contain vitamins, minerals, dietary fiber, and other components that benefit health. In some cases, fortified foods and dietary supplements are useful when it is not possible to meet needs for one or more nutrients (for example, during specific life stages such as pregnancy). For more information about building a healthy dietary pattern, see the [Dietary Guidelines for Americans](#).

Where can I find out more about phosphorus?

- For general information on phosphorus
 - Office of Dietary Supplements (ODS) [Health Professional Fact Sheet on Phosphorus](#)
- For more information on food sources of phosphorus
 - ODS [Health Professional Fact Sheet on Phosphorus](#)
 - USDA's [FoodData Central](#)
- For more advice on choosing dietary supplements
 - ODS [Frequently Asked Questions: Which brand\(s\) of dietary supplements should I purchase?](#)
- For information about building a healthy dietary pattern
 - [Dietary Guidelines for Americans](#)

Glossary

absorption

In nutrition, the process of moving protein, carbohydrates, fats, and other nutrients from the digestive system into the bloodstream. Most absorption occurs in the small intestine.

adverse event

An unwanted medical problem that occurs during treatment. Adverse events may be unrelated to the treatment or they may be caused by the therapy or procedure. For example, an adverse event may be caused by the toxic effects of a particular drug or dietary supplement or by an interaction with another therapy. Also called adverse effect and side effect.

anemia

A condition in which the number of red blood cells in the blood, or the amount of hemoglobin in them, is lower than normal, causing a condition in which red blood cells are not able to supply enough oxygen to all the tissues in the body. Hemoglobin is the substance in red blood cells that carries oxygen to your body's cells.

blood vessel

A tube through which blood circulates in the body. Blood vessels include a network of arteries, arterioles, capillaries, venules, and veins.

calcium

A mineral found throughout the body. Calcium is needed for healthy bones and teeth, for nerves and enzymes to function properly, and for blood clotting. Calcium is found in some foods, including milk, yogurt, and cheese, and in Chinese cabbage, kale, broccoli and fortified foods, such as many drinks, tofu, and cereals.

calcium carbonate

A chemical compound naturally found in chalk, some seashells and other substances. Calcium carbonate is used in antacid drugs to treat indigestion and as a source of calcium to supplement the diet.

cell

The individual unit that makes up the tissues of the body. All living things are made up of one or more cells, which are the smallest units of living structure capable of independent existence.

deficiency

An amount that is not enough; a shortage.

dietary fiber

A substance in plants that you cannot digest. It adds bulk to your diet to make you feel full, helps prevent constipation, and may help lower the risk of heart disease and diabetes. Good sources of dietary fiber include whole grains (such as brown rice, oats, quinoa, bulgur, and popcorn), legumes (such as black beans, garbanzo beans, split peas, and lentils), nuts, seeds, fruit, and vegetables.

Dietary Guidelines for Americans

Advice from the federal government to promote health and reduce the chance (risk) of long-lasting (chronic) diseases through nutrition and physical activity. The Guidelines are updated and published every 5 years by the US Department of Health and Human Services and the US Department of Agriculture.

dietary supplement

A product that is intended to supplement the diet. A dietary supplement contains one or more dietary ingredients (including vitamins, minerals, herbs or other botanicals, amino acids, and other substances) or their components; is intended to be taken by mouth as a pill, capsule, tablet, or liquid; and is identified on the front label of the product as being a dietary supplement.

dose

The amount of medicine or other substance taken at one time or over a specific period of time.

fortified

When nutrients (such as vitamins and minerals) are added to a food product. For example, when calcium is added to orange juice, the orange juice is said to be "fortified with calcium". Similarly, many breakfast cereals are "fortified" with several vitamins and minerals.

gene

The functional and physical unit of heredity passed from parent to offspring. Genes are pieces of DNA, and most genes contain the information for making a specific protein.

genetic disorder

A disease or disorder caused by an alteration or variation (mutation) in a gene or group of genes in the cells of an individual. Examples of genetic disorders include breast cancer, cystic fibrosis, Parkinson's disease, and celiac disease. They can be inherited or can occur without a known cause.

infant

A child younger than 12 months old.

infection

The invasion and spread of germs in the body. The germs may be bacteria, viruses, yeast, or fungi.

interaction

A change in the way a dietary supplement acts in the body when taken with certain other supplements, medicines, or foods, or when taken with certain medical conditions. Interactions may cause the dietary supplement to be more or less effective, or cause effects on the body that are not expected.

kidney

One of two organs that remove waste from the blood (as urine). The kidneys also make erythropoietin (a substance that stimulates red blood cell production) and help regulate blood pressure. The kidneys are located near the back under the lower ribs.

kidney disease

A condition that lessens the ability of the kidneys to filter wastes from the blood, keep blood chemical levels balanced, and make certain hormones. The two most common causes of kidney disease are diabetes and high blood pressure. People with long-term kidney disease may need dialysis or a kidney transplant to stay alive.

laxative

A substance that moves the bowels and relieves constipation.

legume

Dried beans and peas, including kidney beans, pinto beans, black beans, navy beans, lima beans, black-eyed peas, garbanzo beans (chickpeas), split peas, and lentils. Legumes are good sources of protein, iron, zinc, dietary fiber, folate, and potassium.

malnutrition

A disorder caused by a diet that does not provide enough nutrition, an unbalanced diet, a digestive system that does not work properly, or a problem with absorbing or using nutrients.

milligram

mg. A measure of weight. It is a metric unit of mass equal to 0.001 gram (it weighs 28,000 times less than an ounce).

mineral

In nutrition, an inorganic substance found in the earth that is required to maintain health.

multivitamin/mineral dietary supplement

MVM. A product that is meant to supplement the diet. MVMs contain a variety of vitamins and minerals. The number and amounts of these nutrients can vary substantially by product.

nutrient

A chemical compound in food that is used by the body to function and maintain health. Examples of nutrients include proteins, fats, carbohydrates, vitamins, and minerals.

Office of Dietary Supplements

ODS, Office of Disease Prevention, Office of Director, National Institutes of Health, Department of Health and Human Services. ODS strengthens knowledge and understanding of dietary supplements by evaluating scientific information, stimulating and supporting research, disseminating research results, and educating the public to foster an enhanced quality of life and health for the US population.

poultry

Birds that are raised for eggs or meat, including chickens, turkeys, ducks, and geese.

prevent

To stop from happening.

red blood cell

A cell that carries oxygen to and removes carbon dioxide from all parts of the body.

risk

The chance or probability that a harmful event will occur. In health, for example, the chance that someone will develop a disease or condition.

supplement

A nutrient that may be added to the diet to increase the intake of that nutrient. Sometimes used to mean dietary supplement.

upper limit

UL. The largest daily intake of a nutrient considered safe for most people. Taking more than the UL is not recommended and may be harmful. The UL for each nutrient is set by the Food and Nutrition Board at the National Academies of Sciences, Engineering, and Medicine. For example, the UL for

vitamin A is 3,000 micrograms/day. Women who consume more than this amount every day shortly before or during pregnancy have an increased chance (risk) of having a baby with a birth defect. Also called the tolerable upper intake level.

US Department of Agriculture

USDA promotes America's health through food and nutrition, and advances the science of nutrition by monitoring food and nutrient consumption and updating nutrient requirements and food composition data. USDA is responsible for food safety, improving nutrition and health by providing food assistance and nutrition education, expanding markets for agricultural products, managing and protecting US public and private lands, and providing financial programs to improve the economy and quality of rural American life.

vitamin

A nutrient that the body needs in small amounts to function and maintain health. Examples are vitamins A, C, and E.

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